ECEN 454 - Lab Report

Lab Number: 1

Lab Title: Introduction to Cadence Schematic Design & Simulation

Section Number: 341

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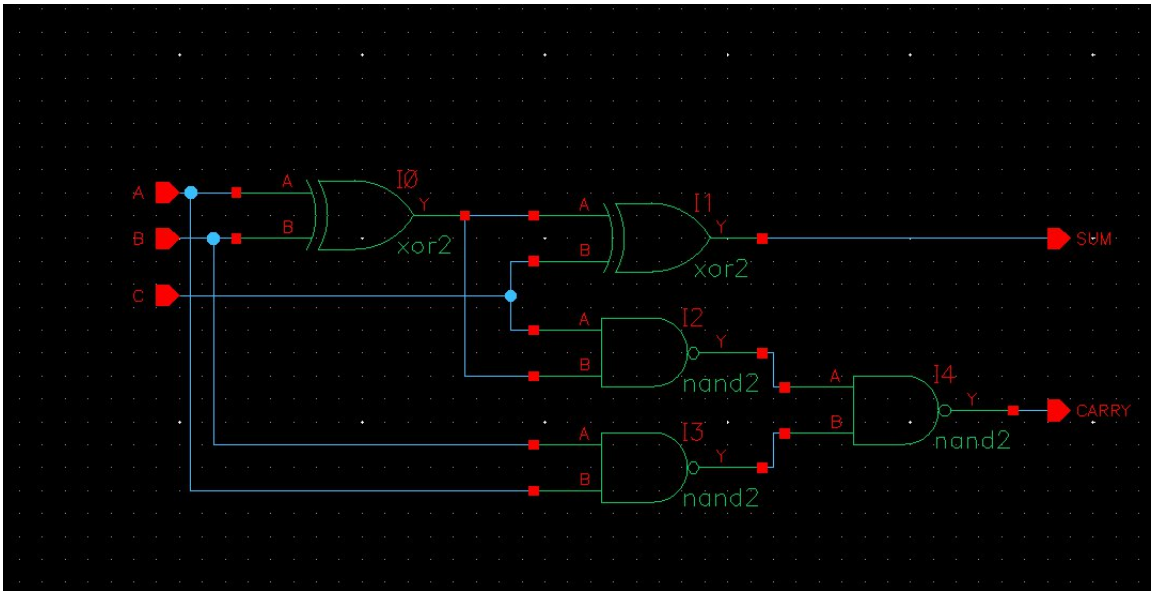
Date: 9/14/2024

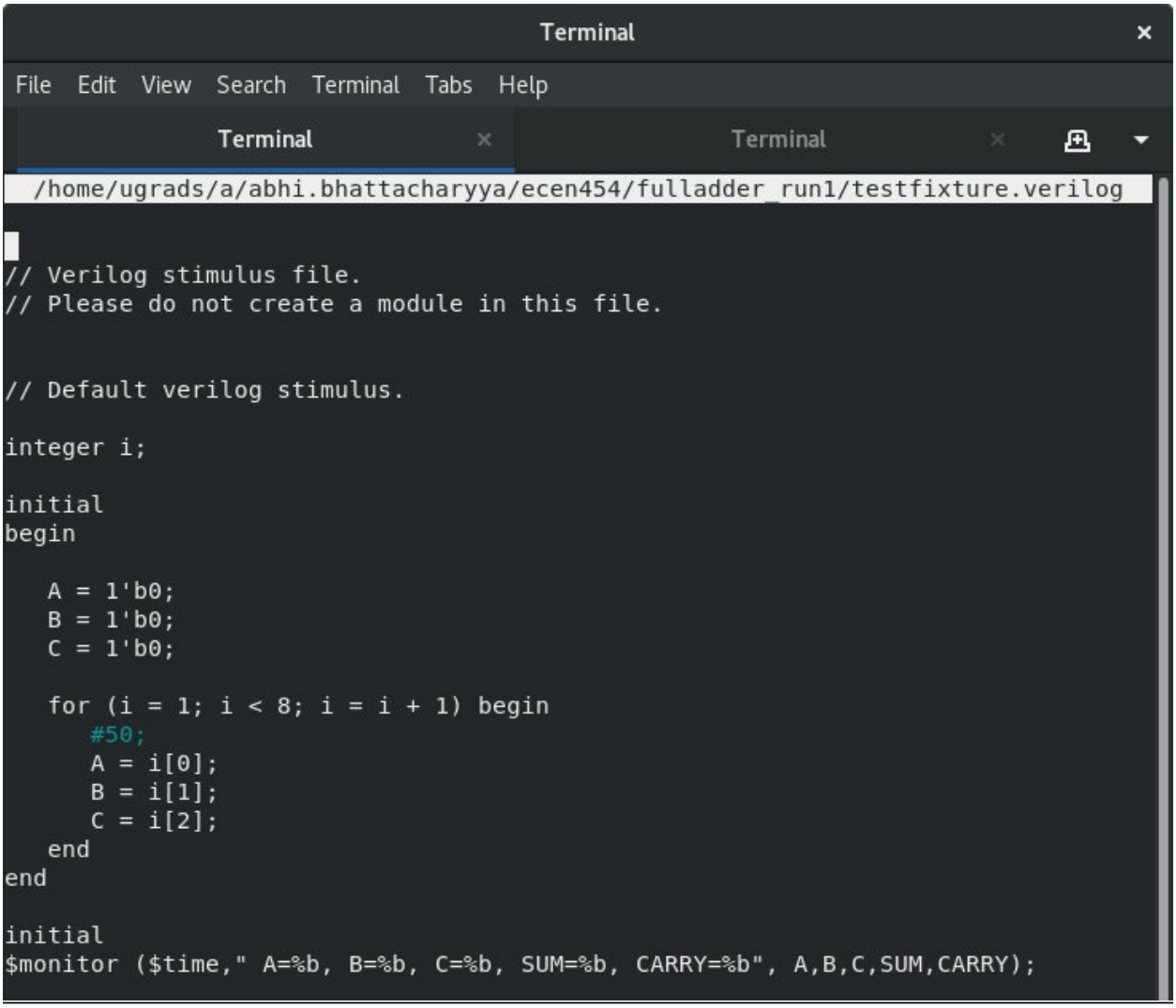
TA: Nishank Bansal

Full Adder Schematic

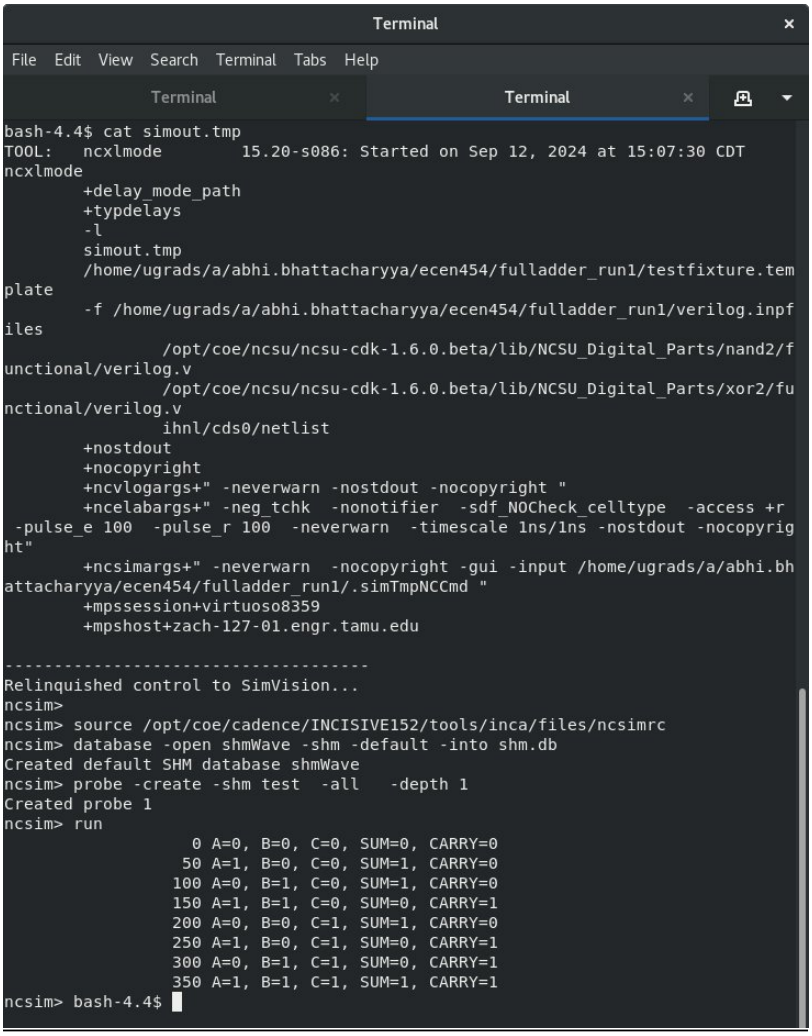


Full Adder Symbol

Full Adder Test Stimulus



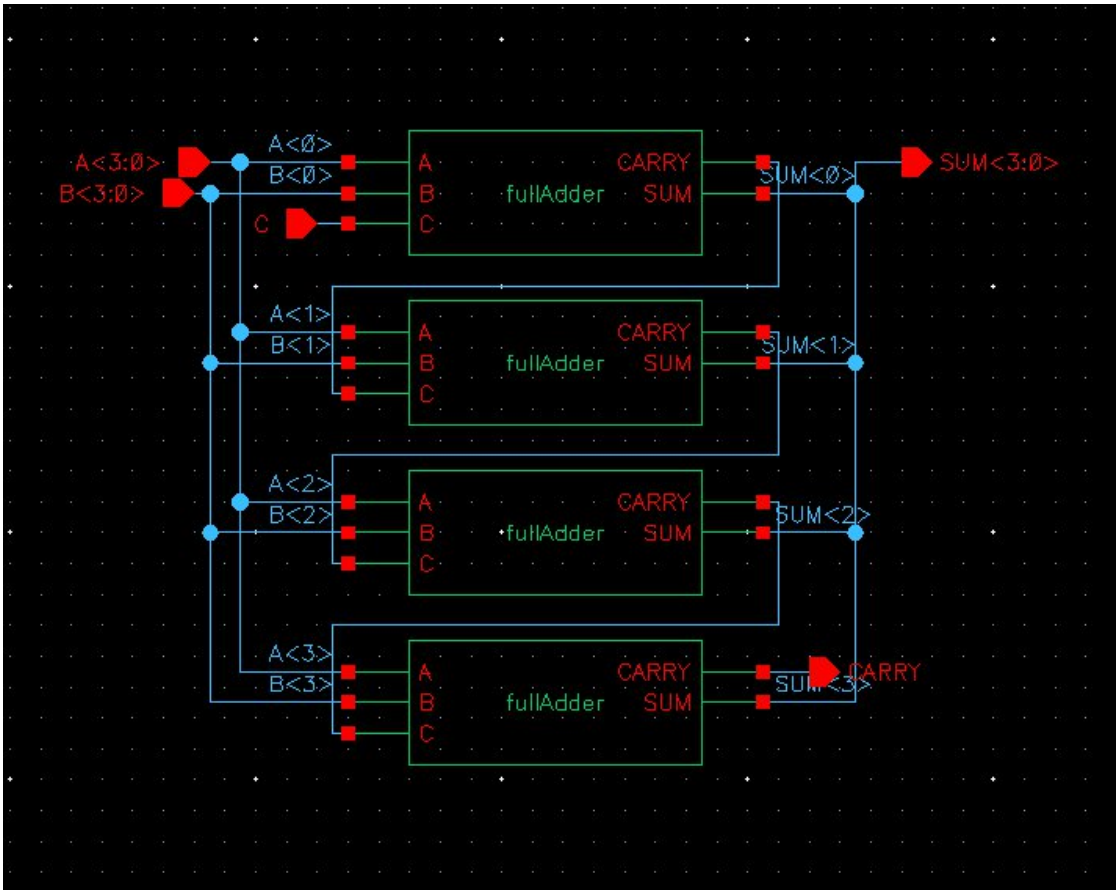
Full Adder Sim Results vs. Expected Results Truth Table



A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

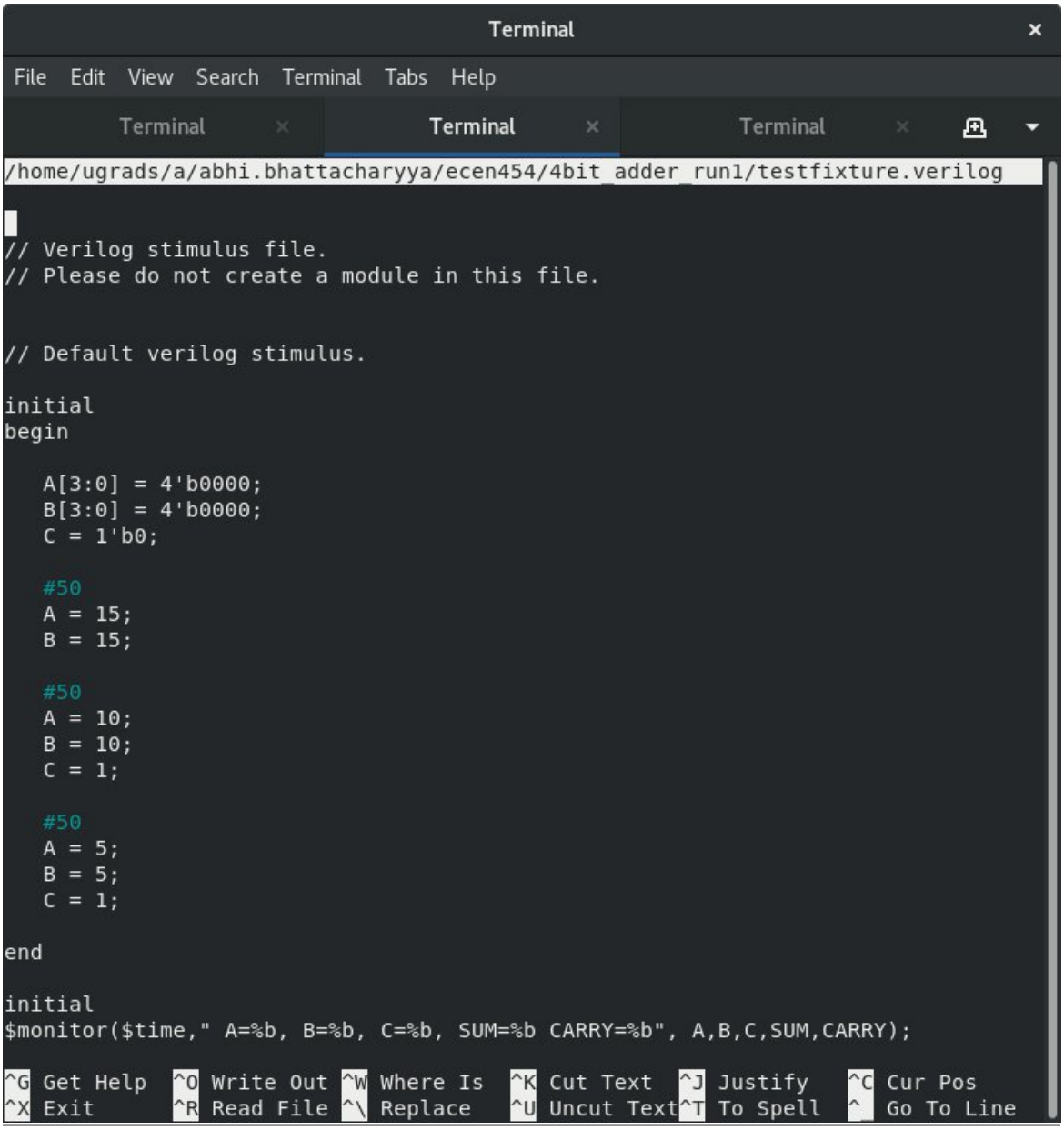
All available input combinations were tested above in the simulation for the full adder. All calculations match the given truth table, so this design passes all tests in the simulation.

4-Bit Adder Schematic

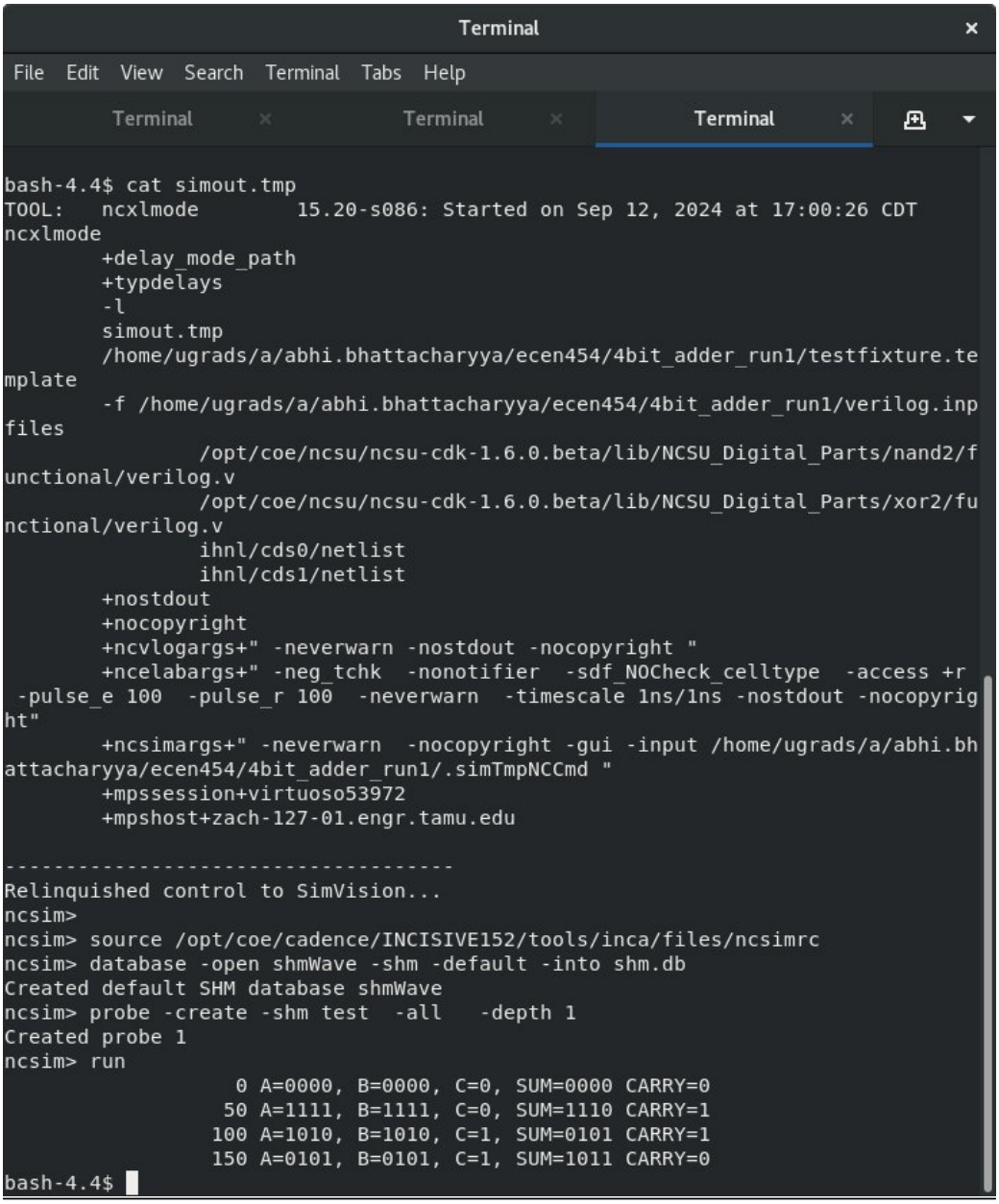


4-bit Adder Symbol

4-bit Adder Test Stimulus



4-bit Adder Sim Results

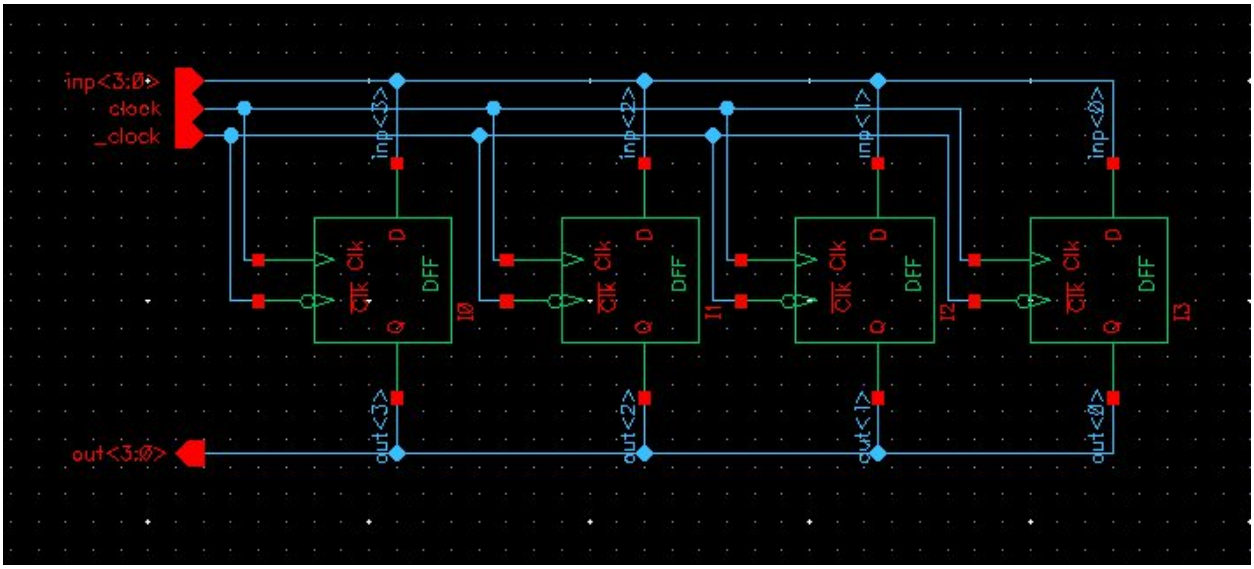


The three specified cases were simulated for the 4-bit adder. The expected results were:

- 1111 + 1111 + 0 = 0000 1
- 1010 + 1010 + 1 = 1110 1
- 0101 + 0101 + 1 = 1011 0

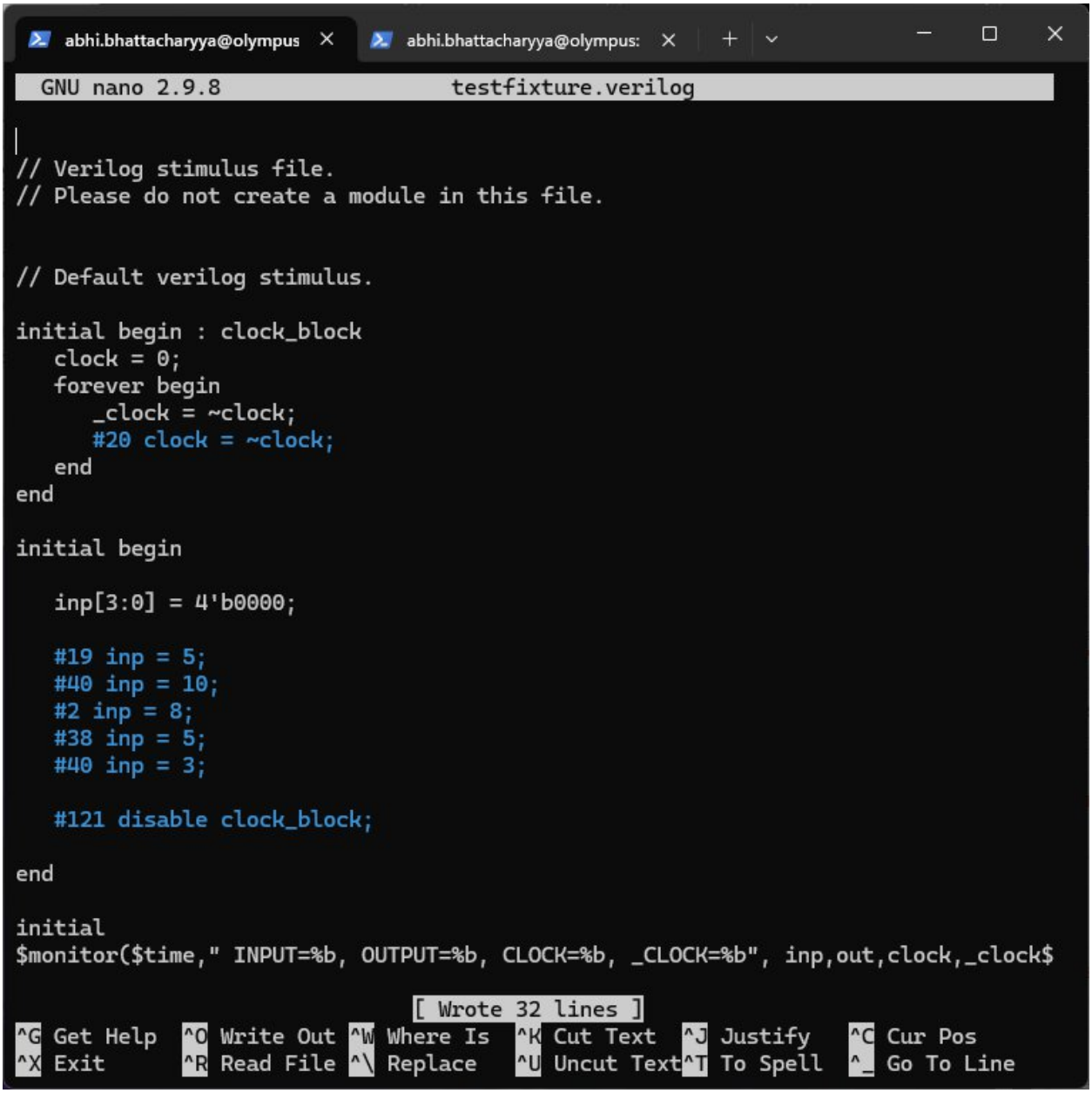
All test results from the simulation match the expected results for these given cases, so the module passes all tests.

4-Bit Register Schematic



4-Bit Register Symbol

4-Bit Register Test Stimulus



4-Bit Register Sim Results

```
ncsim> source /opt/coe/cadence/INCISIVE152/tools/inca/files/ncsimrc
ncsim> database -open shmWave -shm -default -into shm.db
Created default SHM database shmWave
ncsim> probe -create -shm test  -all   -depth 1
Created probe 1
ncsim>
-----------------------------------
Relinquished control to SimVision...
# Restoring simulation environment...
ncsim> input -quiet .reinvoke.sim
ncsim> file delete .reinvoke.sim
ncsim> run
                   0 INPUT=0000, OUTPUT=xxxx, CLOCK=0, _CLOCK=1
                  19 INPUT=0101, OUTPUT=xxxx, CLOCK=0, _CLOCK=1
                  20 INPUT=0101, OUTPUT=0101, CLOCK=1, _CLOCK=0
                  40 INPUT=0101, OUTPUT=0101, CLOCK=0, _CLOCK=1
                  59 INPUT=1010, OUTPUT=0101, CLOCK=0, _CLOCK=1
                  60 INPUT=1010, OUTPUT=1010, CLOCK=1, _CLOCK=0
                  61 INPUT=1000, OUTPUT=1010, CLOCK=1, _CLOCK=0
                  80 INPUT=1000, OUTPUT=1010, CLOCK=0, _CLOCK=1
                  99 INPUT=0101, OUTPUT=1010, CLOCK=0, _CLOCK=1
                 100 INPUT=0101, OUTPUT=0101, CLOCK=1, _CLOCK=0
                 120 INPUT=0101, OUTPUT=0101, CLOCK=0, _CLOCK=1
                 139 INPUT=0011, OUTPUT=0101, CLOCK=0, _CLOCK=1
                 140 INPUT=0011, OUTPUT=0011, CLOCK=1, _CLOCK=0
                 160 INPUT=0011, OUTPUT=0011, CLOCK=0, _CLOCK=1
                 180 INPUT=0011, OUTPUT=0011, CLOCK=1, _CLOCK=0
                 200 INPUT=0011, OUTPUT=0011, CLOCK=0, _CLOCK=1
                 220 INPUT=0011, OUTPUT=0011, CLOCK=1, _CLOCK=0
                 240 INPUT=0011, OUTPUT=0011, CLOCK=0, _CLOCK=1
ncsim> [abhi.bhattacharyya@olympus 4bit_register_run1]$
```

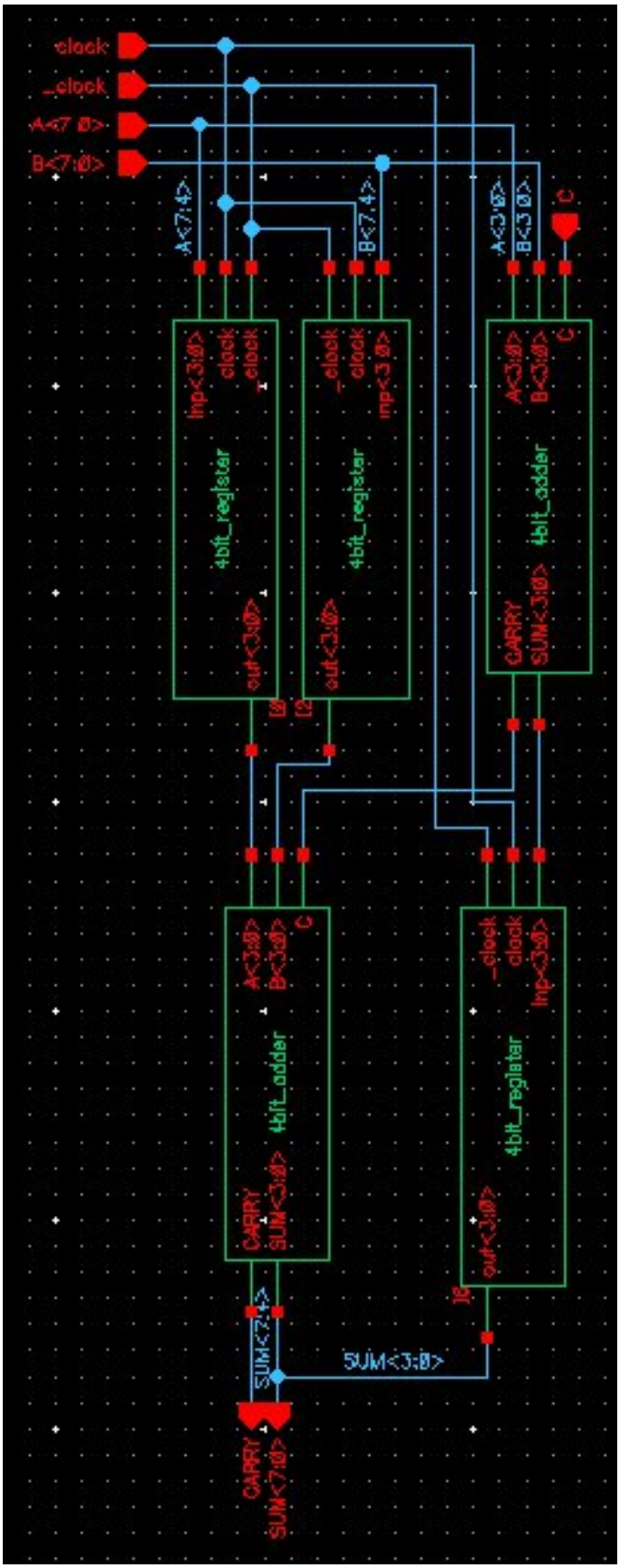
As shown in the test stimulus code, the register was simulated on a clock signal with a period of 40 time units. The first clock rising edge is at t = 20 time units. A race condition was observed when setting the input signal exactly on the rising edge (20 + 40n), where the register's output would sometimes update with the input; and other times remain unchanged. This behavior was random and nondeterministic, pointing to a race condition. To avoid this, the input was changed one time unit before the register's clock cycle.

The cases tested for the 4-bit register were as follows.

Time	Input	Expected Output	
0	0000	XXXX	
19	0101		
20	0101	0101	
59	1010	0101	
60	1010	1010	
61	1000	1010	(Changing input in the middle of a clock cycle to test for issues)
99	0101	1010	
100	0101	0101	
139	0011	0101	
140	0011	0011	

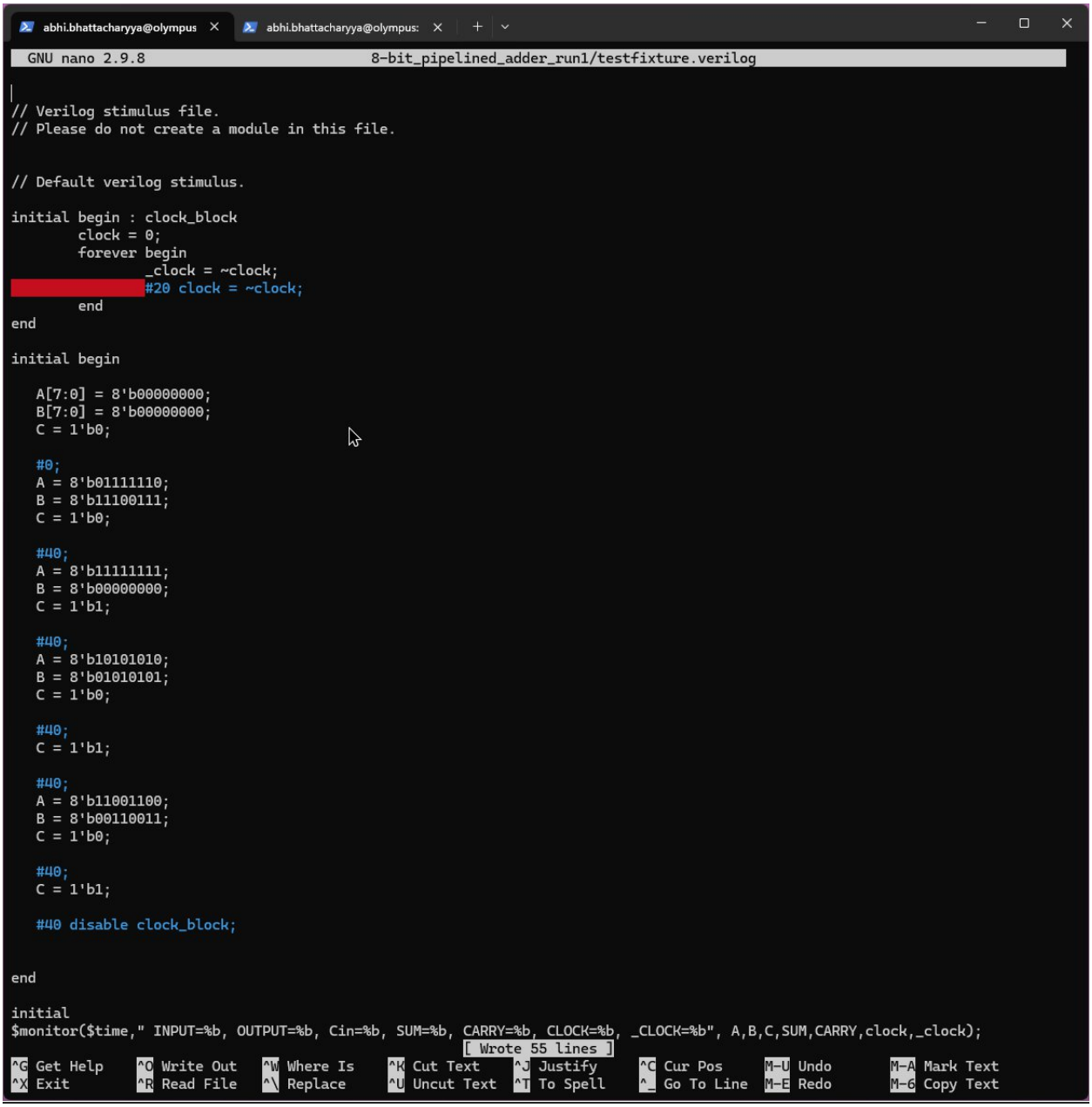
The simulation results matched the expected results, so the register design worked correctly.

8-Bit Pipelined Adder Schematic & Symbol

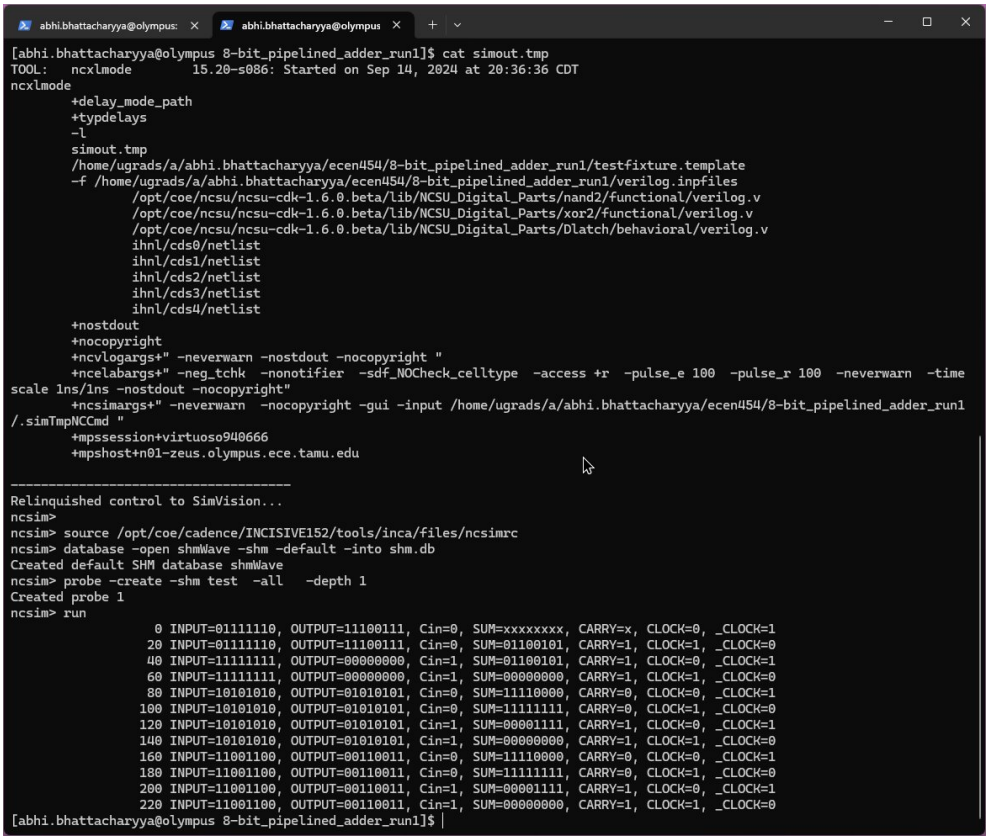

clock
_clock
A<7:0>
B<7:0>
A<7:4>
B<7:4>
A<3:0>
B<3:0>
C
inp<3:0>
4bit_register
out<3:0>
A<3:0>
B<3:0>
4bit_adder
CARRY
SUM<3:0>
SUM<7:4>
SUM<3:0>
CARRY
SUM<7:0>


A<7:0>
B<7:0>
C
_clock
clock
CARRY
SUM<7:0>
8bit_pipelined_adder

8-Bit Pipelined Adder Test Stimulus



8-Bit Pipelined Adder Sim Results



As shown in the test stimulus code, the register was simulated on a clock signal with a period of 40 time units. The first clock rising edge is at t = 20 time units.

The six specified cases were simulated for the 4-bit adder. The expected results were:

Time	A	B	C	SUM	CARRY
20	1111110	11100111	0	01100101	1
60	11111111	0	1	00000000	1
100	10101010	1010101	0	11111111	0
140	10101010	1010101	1	00000000	1
180	11001100	00110011	0	11111111	0
220	11001100	00110011	1	00000000	1

Time units were added above to match the clock period (40) and initialization time (t = 20) used for simulation. The simulation results and calculations match the expected results. One thing of note is that the upper 4 bits of the sum update on the first clock cycle's falling edge, and the lower 4 bits update on the next clock cycle's rising edge. This is due to the layout of registers in the pipelined adder, as the design was optimized to use fewer registers. Since the output of this adder should only be read during each rising clock cycle, and the output results are correctly computed by the time of each rising clock cycle, this design satisfies the specifications for Lab 1.